

AgentForge V2.3 Architecture Hardening & Empirical Validation Audit

Native Antigravity Multi-Agent Software Engineering Architecture

Version: 2.3.0	Date: September 2026	Validation Pass: 100% (8/8 Tests)
Runtime: Antigravity Native	Orchestrator: Parent Session	Audit Status: Hardened & Verified

1. Executive Summary & Audit Context

Following runtime validation across real projects and the controlled OmniCalc full-stack testbench, the AgentForge multi-agent architecture was transitioned from V2.2 to V2.3. While core parallel execution, Playwright CDP verification, and file conflict serialization were proven, the runtime audits revealed three concrete architectural defects requiring formal hardening:

- **Manager Architecture Ambiguity:** Clarifying Parent Antigravity Session as the direct Orchestrator / Manager executing the thin deterministic control loop, preventing nested subagent permission failures.
- **Coder / Tester Test Redundancy:** Coders executing full test suites locally before completion, only for Strict Tester to immediately repeat the identical suite, inflating token costs and step counts.
- **Missing Physical Disk Persistence Gate:** Coder subagents occasionally completing execution with code blocks in conversational text rather than writing to physical disk, causing downstream testers to evaluate missing or stale files.
- **Stitch MCP Operational Truth:** Rigorous classification of Stitch capabilities—confirming API connectivity, local PNG grounding, and token style extensions, while transparently labeling automated text-to-screen generation as unvalidated in the testbench.

2. Architecture Hardening Implementation

Priority 1: Parent-Led Thin Orchestrator & Observability

In the AgentForge runtime, the Parent Antigravity Session acts as the deterministic Manager, dispatching 5 child subagents: context_extractor, planner, coder, strict_tester, and browser_qa. The Manager emits 11 structured observability events (START, DEPENDENCY_CHECK, PREREQUISITE_CHECK, PARALLEL_WAVE, DISPATCH, RECEIVE, VERIFY_DISK_PERSISTENCE, ROUTE strict_tester, ROUTE browser_qa, TASK VERIFIED, NEXT WAVE) and strictly adheres to the Thin Orchestrator Rule (never loading whole files, never synthesizing code for Coder).

Priority 2: Elimination of Coder / Tester Test Redundancy

Coders are now restricted strictly to syntax validation (python -m py_compile, compileall, tsc --noEmit) or minimal sanity checks. Coders are explicitly forbidden from running functional test suites. Strict Tester is established as the sole authoritative runner of functional, targeted, related, and integration test suites in cleanroom isolation, saving an estimated 15k–35k tokens per verified task.

Priority 3: Hard Disk Persistence Gate (Step 3.5)

Before dispatching Strict Tester, the Manager executes Step 3.5: physically verifying on disk all files declared in `modify_files` and `create_files`. If any file is missing, the Manager emits `FAILED_GATE: CODER_OUTPUT`, halts Tester dispatch, formats a compact failure context (<500 tokens), and triggers an immediate Coder retry without synthesizing code on Coder's behalf.

3. Focused Hardening Validation Results (Tests 1–8)

The hardened architecture was evaluated against the OmniCalc system validation environment via `scripts/run_v23_focused_validation.py`. All 8 tests passed with complete empirical evidence logged:

#	Test Name	Target Capability	Status	Evidence Artifact
TEST 1	Manager Observability	11 structured execution event markers verified in Manager specification	PASS	evidence/V2.3-HARDENING/test1_manager_observability.txt
TEST 2	Coder/Tester Separation	Coder syntax check (<code>py_compile</code>); Strict Tester authoritative <code>pytest</code>	PASS	evidence/V2.3-HARDENING/test2_separation.txt
TEST 3	Disk Persistence Gate	Missing file triggers <code>FAILED_GATE: CODER_OUTPUT</code> ; retry verifies disk	PASS	evidence/V2.3-HARDENING/test3_disk_gate.txt
TEST 4	Parallel Concurrency Gate	Disjoint paths concurrent; overlapping paths serialized	PASS	evidence/V2.3-HARDENING/test4_parallel_gates.txt
TEST 5	Browser QA Routing Gate	Backend tasks skip Browser QA; UI tasks enforce Chrome CDP verification	PASS	evidence/V2.3-HARDENING/test5_browser_routing.txt
TEST 6	Human Prerequisite Gate	Missing external API key blocks lane; provisioning unblocks	PASS	evidence/V2.3-HARDENING/test6_human_gate.txt
TEST 7	Stitch Visual Reference Gate	Authoritative visual reference verified on disk; missing reference halts	PASS	evidence/V2.3-HARDENING/test7_stitch_reference.txt
TEST 8	Change Request Lifecycle	Mode B additive change verified; baseline & canonical report preserved	PASS	evidence/V2.3-HARDENING/test8_change_request.txt

4. Stitch MCP Operational Truth & Classification

A rigorous empirical audit of Stitch capabilities was conducted to distinguish working integrations from experimental features:

- **Cloud API Connectivity:** Operational. Stitch MCP is authenticated to Google Cloud and successfully lists cloud design projects (`projects/5578728684898314556`, `projects/4459594925987917940`).
- **Local Visual Grounding:** Operational. Browser QA and Coder inspect authoritative design files (`docs/design/calculator/calculator.png`) to verify color tokens, layout, and button styling.
- **Design Language Token Extension:** Operational. New screens (such as `settings.html`) extract and reuse established design tokens (`--color-accent-equals`, `--radius-card`) without inventing inconsistent styling.
- **Automated Text-to-Screen Generation:** Unvalidated in Testbench. Automated screen generation was not exercised during the OmniCalc run. The system relies safely on local visual assets and design extensions rather than pretending generative creation is fully autonomous.

5. System Maturity Assessment & Known Boundaries

AgentForge V2.3 is a hardened, resilient multi-agent software engineering framework. However, engineering honesty requires stating known operational limits:

- **Production Scope:** V2.3 is validated for controlled full-stack projects. Broad enterprise deployment should occur with human supervision.
- **Process Management on Windows:** Automated spawning and termination of background servers (Uvicorn, Playwright) requires dedicated port hygiene to prevent lingering locks.

- **Transcript Context Limits:** Extensive stdout emissions in subagents are compact-filtered to prevent Parent context exhaustion.

6. Audit Sign-Off & Status

Audit Status: APPROVED & HARDENED	Test Score: 8/8 PASSED (100%)
Specification Version: AgentForge V2.3	Release Tag: v2.3.0-hardening